

A Proposal of an End-to-End DoA Estimation System Aided by Deep Learning

Daniel Akira Ando*, Toshihiko Nishimura*, Takanori Sato*
Takeo Ohgane*, Yasutaka Ogawa*, and Junichiro Hagiwara*

*Graduate School/Faculty of Information Science and Technology

Hokkaido University, Sapporo, Japan

Email: {dakiraando, nishim, tksato, ohgane, ogawa, jhagiwara}@m-icl.ist.hokudai.ac.jp

Abstract—Direction of arrival (DoA) estimation is a technique used in, for instance, radar systems, source localization, and wireless channel estimation, and improving its accuracy is becoming more important as its applications expand. In previous works, we verified that deep neural networks (DNN) trained offline are a viable tool for achieving great on-grid DoA estimation performance, even compared to the traditional root multiple signal classification (root-MUSIC) algorithm. Moreover, we separately proposed the signal-to-noise ratio (SNR)-based DNN selection method in order to fully leverage the DNN learning potential at specific SNRs; and the superposition of two DNNs (staggered DNNs) in order to overcome the estimation failure caused by incident radio waves at the grid border. In this paper, we propose an end-to-end DoA estimation system, which consists of three modules: DoA, SNR and source number estimators. Here, we expand the performance of the DoA estimator module by presenting a new algorithm more efficient in detecting DoA from the DNN output, and by using the combination of the two previously mentioned methods: SNR-based DNN selection and staggered DNNs. Also, an SNR estimation scheme with better precision is presented.

Index Terms—antenna array, DoA estimation, SNR estimation, source number estimation, deep neural network

I. INTRODUCTION

Direction of arrival (DoA) estimation is an extremely important antenna array signal processing technique widely used in several applications, such as radar systems, source localization, and wireless channel estimation. Various techniques have been developed to estimate the DoA of radio waves, such as the MUSIC algorithm, and its extended version: Root-MUSIC [1]. MUSIC is classified as a subspace-based technique, as it makes use of the spectral decomposition of the antenna array received signal correlation matrix. Root-MUSIC is an extension of MUSIC, achieving better estimation performance [1], [2]. However, both techniques require a matrix eigendecomposition computation, which is costly specially when the array dimension increases.

Deep learning techniques applied to communication systems is an ongoing research area. In fact, the applicability of deep learning in the DoA estimation problem has been reported in several studies, such as [3]–[6]. In [3], a framework for end-to-end channel and DoA estimation in the context of massive multiple-input multiple-output (massive MIMO) is proposed.

In [4], a combination of a detection and DoA estimation network, which reduces the training-set size and makes it possible to train several DNNs corresponding to different position sectors, is presented. In [5], a low-complexity DoA estimation technique for hybrid MIMO systems with uniform circular array (UCA) at a base station is presented. In [6], a DoA estimation system which is robust to array imperfections and consists of a multi-task autoencoder and a series of parallel multi-layer classifiers is explained.

Here, we present an end-to-end DoA estimation system aided by DNN. Our system consists of three modules: DoA, SNR, and source number estimators. Compared to our previous studies [9], [10], here we expand the performance of the DoA estimator module (1) by proposing a new algorithm more efficient in retrieving DoA information from the DNN angular grid output, (2) by using the combination of the previously proposed SNR-based DNN selection and staggered DNN methods, and (3) by upgrading the SNR estimator accuracy. Also a simple but powerful learned source number estimator is described.

This paper is structured as follows. The antenna array model considered throughout this paper is explained in Section II. The proposed end-to-end DoA estimation system is explained in Section III. The DNN specialized in DoA estimation, including the method for acquiring training, validation and test datasets, its structure, training procedure and performance improvement strategies is explained in Sections IV and V. SNR estimator module is described in Section VI. The proposed DNN specialized in source number estimation is explained in Section VII. Finally, the performance achieved by our proposed system is compared with Root-MUSIC in Section VIII. Lastly, in Section IX our work is concluded.

II. ANTENNA ARRAY MODEL

A uniform linear array (ULA) with half-wavelength spacing L omnidirectional antenna elements is considered throughout this study. Additionally, no mutual coupling between the antenna elements is assumed. Narrowband radio waves emitted by K sources in the ULA far-field region are impinging on it at angles $\theta = [\theta_1, \dots, \theta_K]^T$, where $(\cdot)^T$ indicates the transpose operator. Then, the baseband received signal $\mathbf{x}(t) \in \mathbb{C}^{L \times 1}$ at the antenna array can be modeled by

$$\mathbf{x}(t) = \mathbf{A}(\theta)\mathbf{s}(t) + \mathbf{z}(t), \quad (1)$$

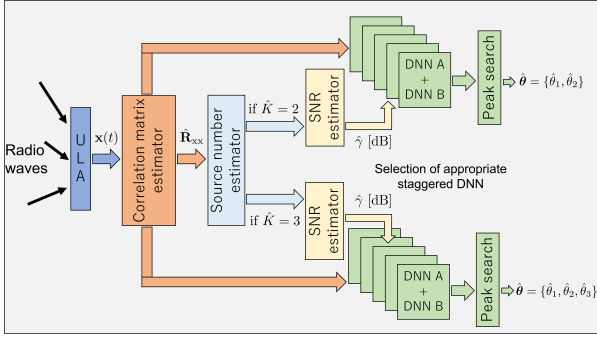


Fig. 1. Proposed system schematics where $\hat{K}$, $\hat{\gamma}$, and $\hat{\theta}$ denote estimated values of the number of sources, SNR, and DoAs, respectively.

where $\mathbf{s}(t) \in \mathbb{C}^{K \times 1}$ is the vector containing the incident radio waves complex amplitudes, $\mathbf{z}(t) \in \mathbb{C}^{L \times 1}$ is the additive white Gaussian noise vector following a circular complex Gaussian distribution $\mathbf{z}(t) \sim \mathcal{CN}(0, \sigma^2 \mathbf{I}_K)$ with zero mean and variance σ^2 , where $\mathbf{I}_K$ represents a K -dimensional identity matrix, and $\mathbf{A}(\theta)$ is the steering matrix, which accounts for the relative delay corresponding to path length difference of the incident waves on each ULA element and is described as

$$\mathbf{A}(\theta) = \begin{bmatrix} 1 & \cdots & 1 \\ e^{-j\pi \sin \theta_1} & \cdots & e^{-j\pi \sin \theta_K} \\ e^{-j\pi 2 \sin \theta_1} & \cdots & e^{-j\pi 2 \sin \theta_K} \\ \vdots & \ddots & \vdots \\ e^{-j\pi(L-1) \sin \theta_1} & \cdots & e^{-j\pi(L-1) \sin \theta_K} \end{bmatrix}. \quad (2)$$

It is also assumed that the incident radio waves are non-coherent and transmitted with equal power normalized to one.

The received signal correlation matrix estimated by N snapshots $\hat{\mathbf{R}}_{xx}$ is given by:

$$\hat{\mathbf{R}}_{xx} = \frac{1}{N} \sum_{n=1}^N \mathbf{x}(t_n) \mathbf{x}(t_n)^H, \quad (3)$$

where $\mathbf{x}(t_n)$ is the n th snapshot taken from the received signal, and $(\cdot)^H$ represents the conjugate transpose operator.

III. END-TO-END DOA ESTIMATION SYSTEM

In this paper, we propose an end-to-end DoA estimation system comprising the following modules:

- DoA estimator (Sections IV and V),
- SNR estimator (Section VI), and
- Source number estimator (Section VII).

The schematics of the proposed system can be seen in Fig. 1. Here it is assumed that either two or three radio waves are impinging on the antenna array at angles at least 1° mutually separated. The correlation matrix is then estimated from N snapshots of the received signal $\mathbf{x}(t)$ at the ULA, as it was explained in (3). For the next three components in the schematics, the correlation matrix is used as an input in order to estimate the respective quantities.

In the first component, the number of radio waves is estimated with an offline trained neural network. Then SNR

estimation is performed. As it will be explained later, just discrete values of SNR are considered, in fact 0, 5, 10, 15 and 30 dB. Lastly, according to the estimated SNR, the appropriate DNNs combination (which are trained offline in advance for each of the discrete SNRs) is selected in order to produce an angle spectrum, on which a DoA detection algorithm is applied. All DoA estimator DNNs are designed with the same architecture, which eases the processing, since only shifting of trained weights between selected DNNs is required.

IV. DOA ESTIMATOR

A. Training Data Preparation

Inputs to fully connected DNNs are generally one dimensional, and forward and backpropagation computation is usually defined in the real domain. For this reason, the estimated correlation matrix $\hat{\mathbf{R}}_{xx}$ must first be modified in order to be used as input data. Noting that $\hat{\mathbf{R}}_{xx}$ is hermitian and thus it can be written as in (4), a properly shaped input vector $\mathbf{u}$ can then be generated as follows: first we arrange the diagonal elements of the correlation matrix as the first entries to the input vector; then, since the lower triangular elements are the complex conjugate of the upper triangular elements, we choose to use only the lower triangular elements, by taking each element real and imaginary part column by column and from left to right and arranging it in the remaining space of the input vector (See (5)).

$$\hat{\mathbf{R}}_{xx} = \begin{bmatrix} r_{11} & r_{21}^* & \cdots & r_{L1}^* \\ r_{21} & r_{22} & \cdots & r_{L2}^* \\ \vdots & \vdots & \ddots & \vdots \\ r_{L1} & r_{L2} & \cdots & r_{LL} \end{bmatrix} \quad (4)$$

$$\mathbf{u} = [r_{11}, \dots, r_{LL}, \Re(r_{21}), \Im(r_{21}), \dots, \Re(r_{L1}), \Im(r_{L1}), \dots, \Re(r_{L(L-1)}), \Im(r_{L(L-1)})]^T. \quad (5)$$

By doing so, an input vector $\mathbf{u}$ with dimension L^2 is acquired and then can be used as input to the DNN.

Next, the angle spectrum is discretized in a finite number of angle bins. As it will be shown in (6) below, each bin represents the probability that a radio wave is incident on it. If so, then the bin should produce an output 1, otherwise 0.

$$y_j = \begin{cases} 1 & \text{if wave is incident on the } j\text{th bin} \\ 0 & \text{otherwise} \end{cases}. \quad (6)$$

Finally the training, validation, and test dataset can be generated by the method described above, and then they can be fed to the DoA estimator DNN for training. Parameters for data generation are summarized in Table I. Although random datasets are simply used in this paper, thorough verification of the above datasets by means of, for instance, K-fold cross validation should be addressed in the future study.

B. DNN Model Training

A standard fully connected DNN is trained and used for DoA estimation. The input to this DNN has in total L^2 units (5), and the output $\{(y_{\max} - y_{\min})/y_{\text{step}} + 1\}$ units,

TABLE I
PARAMETERS FOR DATA GENERATION

Number of antenna elements L	5
Number of incident radio waves K	2, 3
Direction of arrival angles θ	random from -60.5° to $+60.5^\circ$
Number of snapshots N	100
Number of training data samples	250,000
Number of validation data samples	10,000
Number of test data samples	150,000

TABLE II
PARAMETERS FOR DNN TRAINING

Input layer units	25
Hidden layers	4
Units per hidden layer	DNN-A: 363 DNN-B: 366
Output layer units	DNN-A: 121 DNN-B: 122
Hidden layers activation function	ReLU
Output layer activation function	Sigmoid
Loss function	MSE
Optimizer	Adam
Batch size	256
Number of epochs	300

where each unit represents an angle bin (6). In this study, we consider an angle grid from -60° to $+60^\circ$ in steps of 1° , totaling 121 output units. From previous studies results [9], here the number of hidden layers was fixed to four, and each layer has 363 units (366 units in case of DNN-B, explained in Section V-B). This structure is not necessarily the most optimal, but nevertheless good results were obtained in the past empirical evaluation.

Batch normalization, a powerful regularizer normally used as an improvement of the overall stability of the learning process, is also used in all layers [7]. The activation function choice of all hidden layers is the rectified linear unit (ReLU). The output layer activation function is set up as the Sigmoid function. Its use guarantees that each entry in the output vector behaves as probability quantities (6). Mean squared error (MSE) is selected as the loss function. The optimization algorithm used here is Adam [8].

The learning progress is verified with the success rate metric. An estimation success is only counted when all DoAs in a sample were correctly detected. In addition, the model weights corresponding to the best success rate were saved. It is important to notice here that the saved weights are not necessarily optimal in terms of the root mean squared error (RMSE), which is a metric used during the testing stage, and can be expressed as:

$$\text{RMSE} = \sqrt{\frac{1}{KN_t} \sum_{k=1}^K \sum_{n=1}^{N_t} \left(\hat{\theta}_k^{(n)} - \theta_k^{(n)} \right)^2}, \quad (7)$$

where N_t is the total number of test samples.

The DNN training parameters are summarized in Table II.

V. DOA ESTIMATOR PERFORMANCE IMPROVEMENT STRATEGIES

A. Neighbors Average Method

In [9], [10], DoA detection from the DNN output was performed by simply choosing the angle bins corresponding to largest K angle spectrum values (here called “Largest Search” method). Specially when $K = 3$, this method fails to give good detection accuracy, because (1) the DNN produces spurious output values in the neighborhood of the bins corresponding

to true DoAs and (2) at times these spurious values are larger than the true bin values. Applying a peak search algorithm has also not proven to be a good solution. For this reason, we propose a new DoA detection algorithm, here named “Neighbors Average”. The idea is to first search for all largely enough excited bins, and then averaging out mutually near bins. So far only two neighbor bins have been considered, yet it was confirmed that performance has improved considerably, specially when $K = 3$ (See Fig. 2).

B. SNR-based Switching Staggered DNN

As it was explained in [9], radio waves impinging on the angle bin border cause the DNN to concurrently excite neighboring bins, resulting in frequent incorrect DoA detection. The proposed strategy in [9], then named staggered DNNs, was to stack up on top of a main DNN (called DNN-A) a second one (called DNN-B), whose output grid had been shifted a half DNN-A grid spacing, ranging from -60.5° to $+60.5^\circ$. After applying a peak search algorithm¹ to the resultant combined angular grid, it was shown that DoA estimation accuracy increased considerably.

In [10], using specific DNNs trained at different SNRs to detect DoA at the corresponding SNRs was proven to be very effective. This solution consists of estimating the SNR from the correlation matrix and then selecting the appropriate DNN-A to perform DoA detection. The possible SNR estimated values are within the $\{0, 5, 10, 15, 30\}^2$ discrete set.

Here we integrate both ideas: selecting staggered DNNs (DNN-A + DNN-B) based on the estimated value of SNR within the set $\{0, 5, 10, 15, 30\}$. The SNR estimation scheme is discussed in the next section.

VI. SNR ESTIMATOR

In previous studies [10], SNR estimation was performed by training a DNN specific for this task. However, estimation

¹Note that “Neighbors Average” is applicable to the staggered DNN as well as to the single DNN case. However, peak search is used here in conformity with the previous research [9].

²It was shown in [10] that the DNN-A trained at 30 dB has better performance when DoA-detecting at 20 and 25 dB. Therefore complexity is reduced as training DNNs at these SNRs becomes unnecessary.

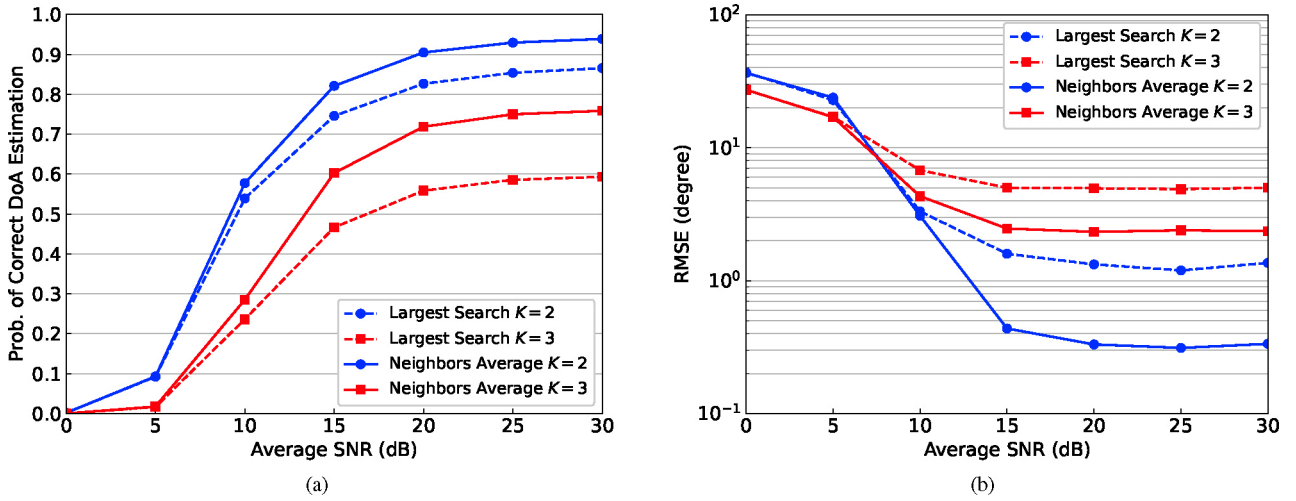


Fig. 2. Estimation accuracy improvement by the proposed technique named “Neighbors Average”. A DNN trained offline with a dataset generated at 30 dB by the parameters at Table I was tested at other SNRs. (a) Probability of correct estimation. (b) RMSE.

accuracy was found to be insufficient. For this reason, here we propose a more accurate strategy for SNR estimation from the estimated correlation matrix. It was verified that the SNR in dB and $-\log(\lambda_s)$ are linearly correlated, where λ_s represents the smallest eigenvalue of the correlation matrix $\hat{\mathbf{R}}_{xx}$. Via standard linear regression, the following estimation function for the SNR when $K = 2$ is obtained:

$$\text{SNR [dB]} = -0.99644 + 9.98865(-\log(\lambda_s)). \quad (8)$$

When $K = 3$:

$$\text{SNR [dB]} = -0.72178 + 10.01046(-\log(\lambda_s)). \quad (9)$$

As explained in Section V, here only discrete values of SNR are considered: 0, 5, 10, 15 and 30 dB. As a result, approximately 94% estimation accuracy has been achieved, largely surpassing the previous 78.8% [10].

The closest value within the set $\{0, 5, 10, 15, 30\}$ to the SNR estimated by (8) and (9) is chosen as the final SNR estimate $\hat{\gamma}$ (See Fig. 1).

VII. SOURCE NUMBER ESTIMATOR

A. Training Data Preparation

The input to this DNN is also the vectorized form of the correlation matrix (See (5)). Since only the cases where two or three emitting sources are considered, the DNN output will be in one-hot representation. This problem can then be viewed as a classification one.

As it was explained in Section III, the first step of the proposed end-to-end system is source number estimation. Consequently, this neural network should be as general as possible to be able to correctly estimate the number of incoming radio waves impinging on the ULA at different SNRs. So we generate training samples by sampling the SNR from a uniform distribution from 0 to 30 dB, and randomly shifting the number of sources between two and three at each sample.

The remaining parameters are kept unchanged as it is detailed in Table I.

B. DNN Model Training

Since there are only two possible classes in the present source number estimation problem, the neural network size is relatively small. The input layer has L^2 units, the output layer has two units, and there are two hidden layers with 50 units each.

The hidden layers and the output layers activation function are ReLU and Softmax, respectively. The choice on the Softmax function lies on the classification nature of this neural network. This also results in choosing the categorical cross-entropy function as the loss function to be minimized.

Batch size is 128, and the total number of epochs is 10. The remaining hyper-parameters are the same as the DoA estimator DNN ones.

After training, testing of this neural network showed that approximately 99.8% of source number estimation accuracy has been achieved for SNR values of 0, 5, 10, 15, 20, 25, and 30 dB. This shows that this DNN has a really good generalized performance.

VIII. PROPOSED SYSTEM PERFORMANCE AND DISCUSSION

The proposed system performance is validated for all methods described in Fig. 3. There, perfect source number estimation is assumed. In terms of the probability of correct DoA estimation (Figs. 3(a) and 3(c)), the proposed SNR-based Switching Staggered DNNs method achieved the best precision. However, in terms of RMSE, it only achieved the best precision in Fig. 3(d) because Staggered DNNs methods use a peak search algorithm according to the previous research [9]. It is expected that applying “Neighbors Average” will improve the RMSE performance of Staggered DNNs methods.

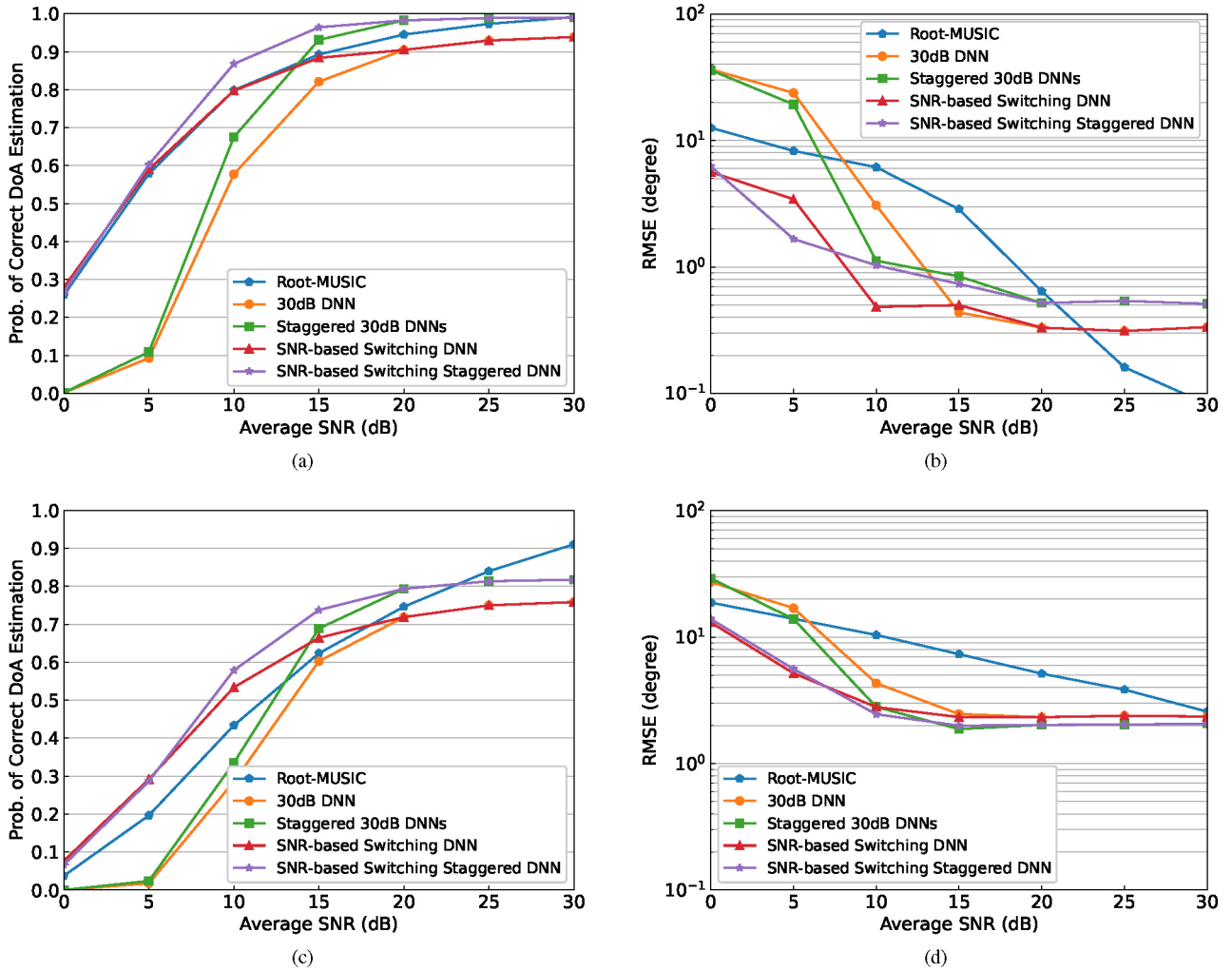


Fig. 3. Metrics comparison from test dataset between the four proposed DoA estimation methods [10] and Root-MUSIC. “30 dB DNN” corresponds to a DNN trained with a dataset generated from 30 dB data; “30 dB Staggered DNN” corresponds to DNN-A and DNN-B trained with a dataset generated from 30 dB data; “SNR-based Switching DNN” corresponds to the method of selecting the appropriate DNN based on SNR estimation; “SNR-based Switching Staggered DNN” is the method combining SNR-based Switching and Staggered DNNs. (a) Probability of correct DoA estimation when $K = 2$. (b) RMSE when $K = 2$. (c) Probability of correct DoA estimation when $K = 3$. (d) RMSE when $K = 3$.

In Fig. 4, the proposed system performance degradation is compared under the conditions detailed in the same figure. While RMSE degradation is evident even when source number estimation is almost perfect (Fig. 4(b)), probability of correct DoA estimation remains virtually unchanged (Fig. 4(a)). It can be said that the RMSE was highly degraded by large estimation errors due to a few source number estimation errors.

A. DNN Output at Incorrect Source Number Detection

It is also important to verify the DoA estimator output in case the number of sources was incorrectly detected. In Fig. 5, the staggered DNN output is shown when the SNR is 20 dB and the two sources were mistaken as three. Likewise, in Fig. 6, the staggered DNN output is shown when the SNR is 20 dB and the three sources were mistaken as two.

As it can be seen first in the situation portrayed in Fig. 5, several angle bins were excited, including the ones corre-

sponding to the true DoAs (shown in green arrows). In fact, correct DoA estimation was achieved for the two true sources. But also a spurious bin was mistakenly taken for the third nonexistent source DoA. In the situation portrayed in Fig. 6, less spurious bins were excited. Also, if the source number of three were estimated, all three true DoAs could have been correctly detected, but only two were output by the estimator.

IX. FINAL REMARKS

In this paper, we have proposed an end-to-end DoA estimation method consisting of DoA, SNR, and source number estimation modules. The DoA estimation module performance has been improved by a new DoA detection method, and by a combination of staggered DNNs and selection of appropriate DNN based on SNR estimation. Additionally, the SNR estimation module has also been significantly enhanced, achieving approximately 94% estimation accuracy under the

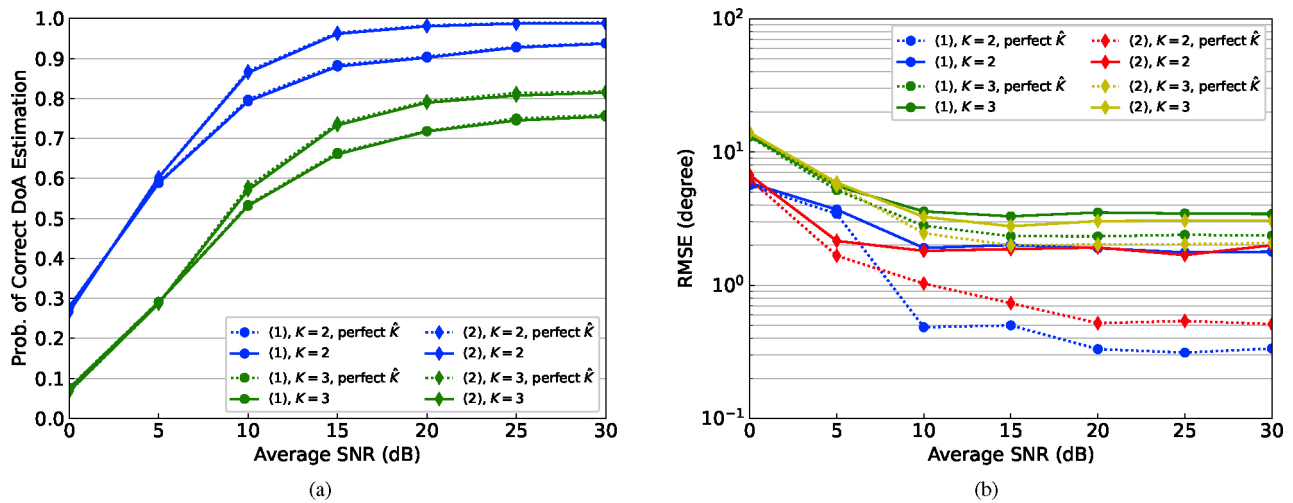


Fig. 4. Proposed system performance comparison when source number estimation is perfect (perfect $\hat{K}$) and not perfect, and when the DoA estimator is (1) Switching DNN or (2) Switching Staggered DNNs. (a) Probability of correct DoA estimation. (b) RMSE.

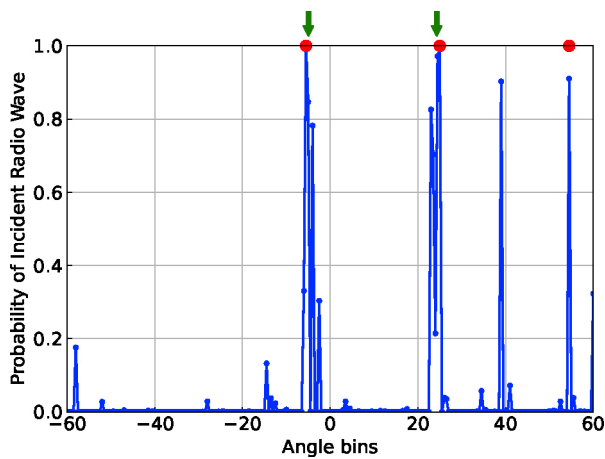


Fig. 5. Staggered DNN output when two sources were mistaken for three. Green arrows represent true DoAs; Red circles, estimated DoAs.

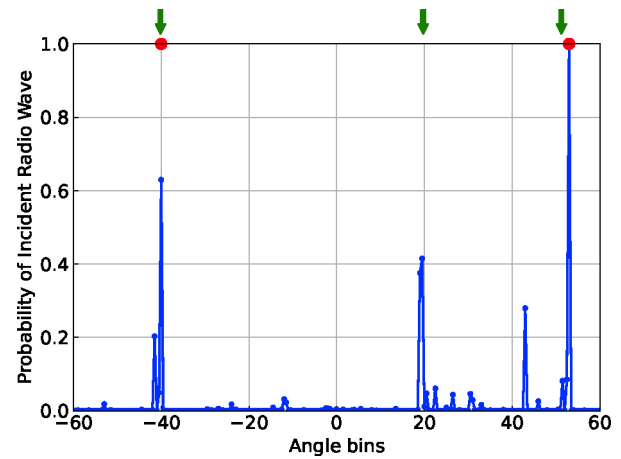


Fig. 6. Staggered DNN output when three sources were mistaken for two. Green arrows represent true DoAs; Red circles, estimated DoAs.

circumstances described here. Lastly, the concept of a new DNN trained specifically for source number estimation has also been presented.

REFERENCES

- [1] H. Krim and M. Viberg, "Two Decades of Array Signal Processing Research: the Parametric Approach," *IEEE Sig. Process. Mag.*, vol. 13, no. 4, pp. 67–94, July 1996.
- [2] B. D. Rao and K. V. S. Hari, "Performance Analysis of Root-MUSIC," *IEEE Trans. Acoust., Speech, Signal Process.*, vol. 37, no. 12, pp. 1939–1949, Dec. 1989.
- [3] H. Huang, J. Yang, H. Huang, Y. Song and G. Gui, "Deep Learning for Super-Resolution Channel Estimation and DOA Estimation Based Massive MIMO System," *IEEE Trans. Veh. Technol.*, vol. 67, no. 9, pp. 8549–8560, Sept. 2018.
- [4] M. Chen, Y. Gong, and X. Mao, "Deep Neural Network for Estimation of Direction of Arrival with Antenna Array," *IEEE Access*, vol. 8, pp. 140688–140698, Aug. 2020.
- [5] D. Hu, Y. Zhang, L. He, and J. Wu, "Low-Complexity Deep-Learning-Based DOA Estimation for Hybrid Massive MIMO Systems with Uniform Circular Arrays," *IEEE Wireless Commun. Letters*, vol. 9, no. 1, pp. 83–86, Jan. 2020.
- [6] Z.-M. Liu, C. Zhang, and P.S. Yu, "Direction-of-Arrival Estimation Based on Deep Neural Networks with Robustness to Array Imperfections," *IEEE Trans. Antennas Propag.*, vol. 66, no. 12, pp. 7315–7327, Dec. 2018.
- [7] S. Ioffe and C. Szegedy, "Batch Normalization: Accelerating Deep Network Training by Reducing Internal Covariate Shift," arXiv:1502.03167v3, Mar 2015.
- [8] D. P. Kingma and J. L. Ba, "Adam: A Method for Stochastic Optimization," arXiv:1412.6980v9, Jan. 2017.
- [9] Y. Kase, T. Nishimura, T. Ohgane, Y. Ogawa, T. Sato, and Y. Kishiyama, "Accuracy Improvement in DOA Estimation with Deep Learning," *IEICE Trans. Commun.*, vol. E105-B, no. 5, pp. 588–599, May 2022.
- [10] Y. Kase, "DOA Estimation of Radio Signals with Deep Learning," Ph.D dissertation, Grad. School of Information Science and Technology, Hokkaido University, Japan, 2022.